Scaling Muon for LLM Training

Deep Dive into Kimi K3 — Chapter 12

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Foundation Books Project

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Selected hidden matrices get an orthogonalized, shape-scaled step;
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Goal: separate the exact polar reference from the finite five-step map, keep decay and shape scale as distinct coefficients, assign optimizers by role, preserve the full matrix under ZeRO-1, and read the 52% claim within its scope.

Roadmap

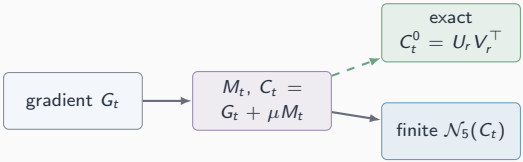
The matrix direction

Scale, decay, and roles

Distribution and evidence

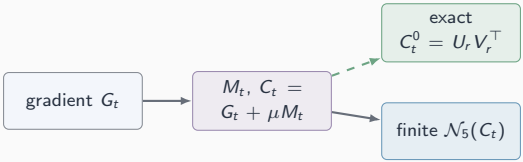
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- The five-step quintic $p(x) = 3.4445x - 4.7750x^3 + 2.0315x^5$ is tuned for a short horizon: $p(1) = 0.701$, and its error is nonmonotone in K. [OBSERVED: companion FP64 probe]

Scale, decay, and roles

One complete hybrid step

$$D_t = c_0 \sqrt{\max(m, n)}\, \mathcal{N}_5(C_t), \qquad W_t = (1 - \eta_t \lambda) W_{t-1} - \eta_t D_t$$

- $\mathrm{RMS}(C^0) = \sqrt{r/(mn)}$: the shape factor equalizes RMS to $c_0 = 0.2$ only at full rank; 0.2 itself is empirical. [REPORTED: Moonlight Sec. 2.2]

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- Role decides the owner: hidden attention, MLP, and router matrices go to Muon; embeddings, heads, norms, biases to AdamW. Running ledger: 315 logical scalars, committed atomically.

Distribution and evidence

Whole matrices, scoped claims

- ZeRO-1 shards state, not the matrix function: gather the ordered candidate, transform once, slice. Independent shard transforms differ by more than 10^{-3}. [OBSERVED: companion fixture]

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Diagnose in dependency order: roles, mathematics, execution, workload. Next: Chapter 13's MuonClip and Training Stability.